



INF 451 - Project Management IS

Fall 2026

Practice 1

Plan

- Getting to know each other
 - Syllabus
 - Introduction
 - Questions and Answers
- 
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Getting to know each other

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Syllabus

Syllabus:

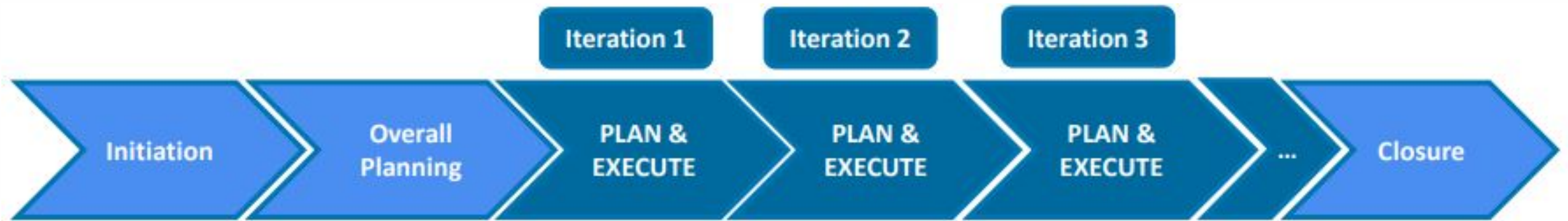
<https://drive.google.com/drive/folders/17isFKpsD6FCw18yh9Gy3O00PRB2-SfPu>

- [Practice lesson](#)
- [Grading sheet](#)
- [Google form](#)



AGILE Project Management

In the past 20-30 years, project complexity has increased significantly. Internet, new technology and globalization have changed the business world. Projects and project management have inevitable been also impacted. The project goals and furthermore their scope have become much more difficult to describe in advance. Hence, planning with a great level of detail, without making multiple changes to this plan is almost impossible. AGILE has been introduced as a new approach, addressing this challenge. The method suggests to plan and execute pieces of the scope and factor any changes in between these pieces. They are called “Iterations” and include detailed planning and the respective execution.

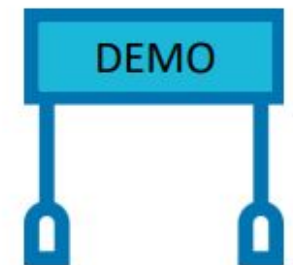


Project Structure

- The main phases in an Scrum projects are standardized “SPRINTS”
- The sprints are of the same duration, usually between 1 and 4 weeks
- Each sprint has to deliver specific part of the product backlog

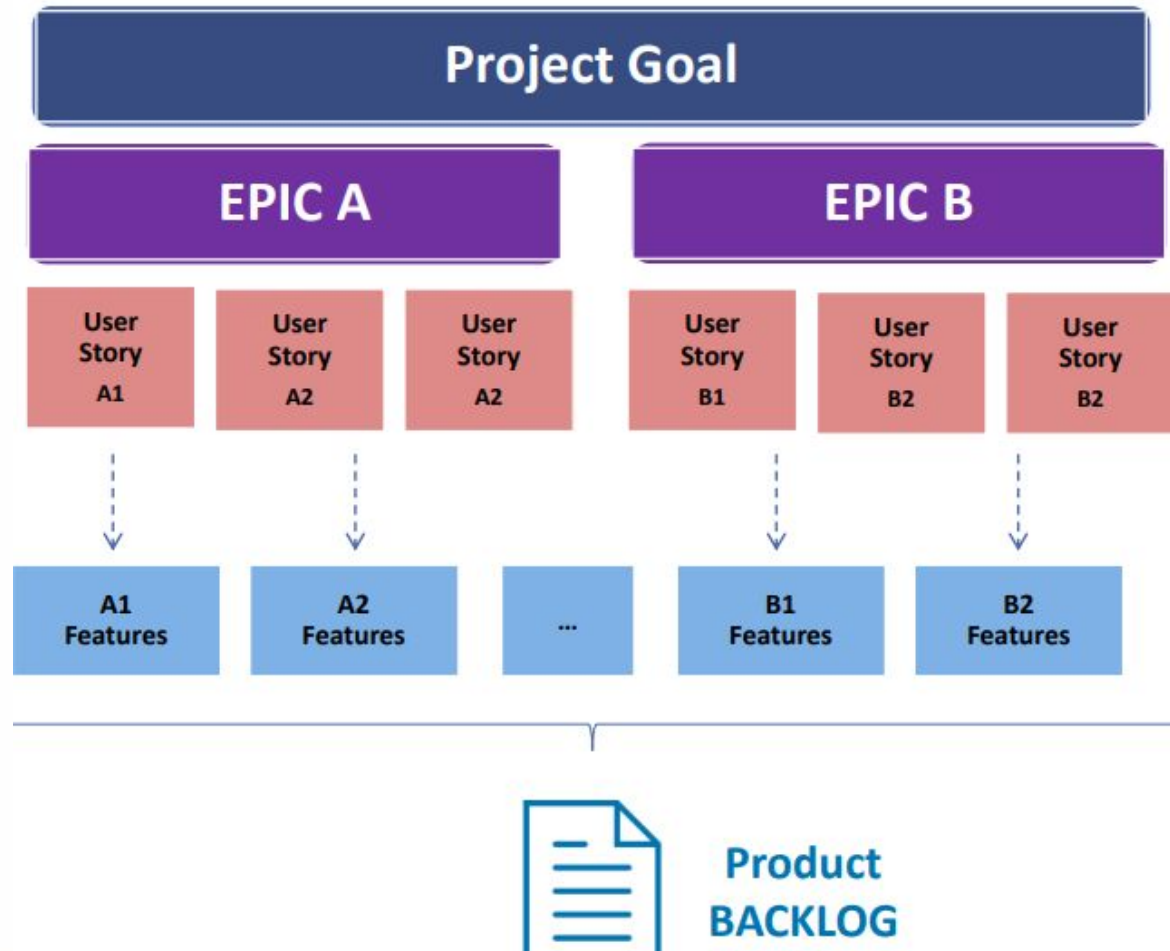


- A “Demo” **D** meeting is organized at the end of each sprint. The goal is to show the completed work to the product owner and stakeholders



Defining the Scope

Defining the Scope:



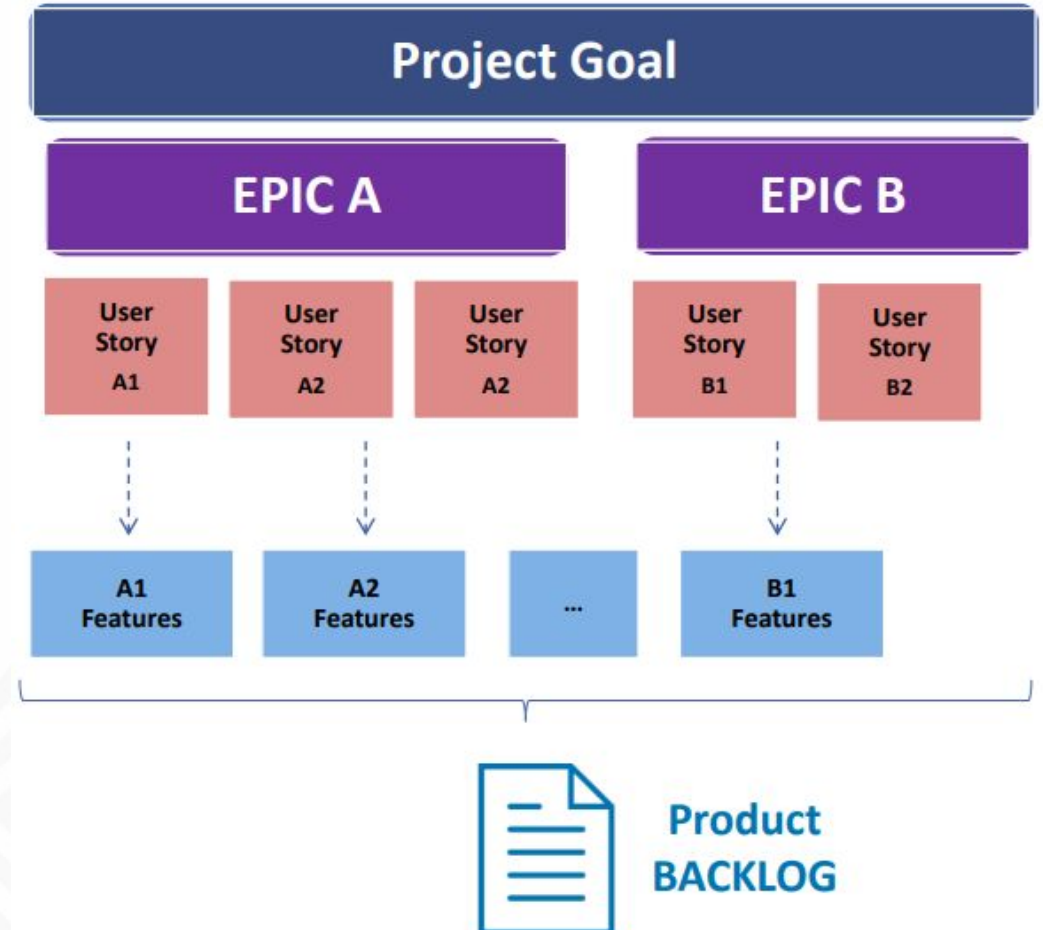
User Story
(format)

**As a user I
want to be
able to ...**

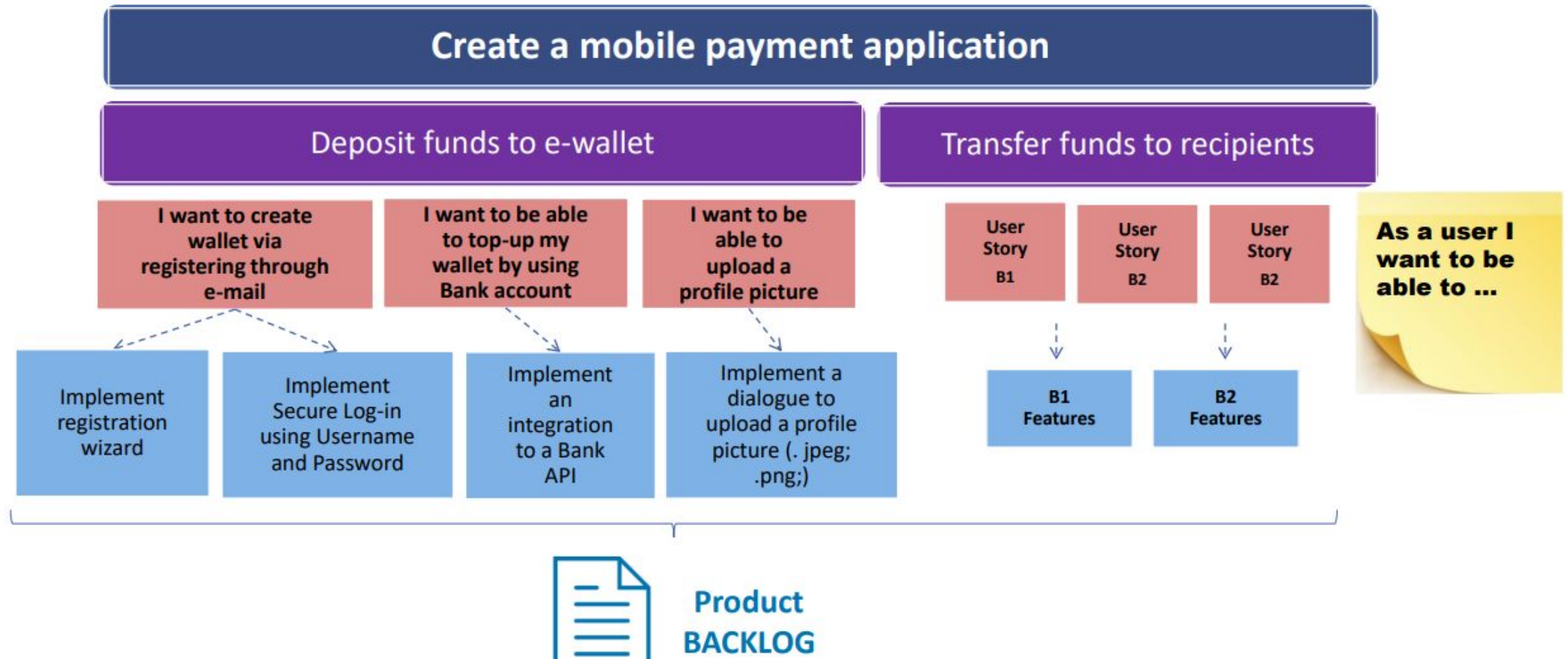
User Story	Feature
I want to create wallet via registering through e-mail;	- Implement registration wizard for first time users - Implement Secure Log-in using Username and Password - Implement Log-out
I want to be able to upload a profile picture	Implement a dialogue to upload a profile picture (. jpeg; .png;)

Note: 1 user story may correspond to one or more Features

Finally, all Features and User Stories are collected in a list. This list is the backbone of the project Scope and is called “Product Backlog”



Defining the Scope: Example



SCRUM: Concepts

Defining the Scope:

The scope is structured from high to low level, in the following way:

- As always, you start from the Project Goal definition: E.g. As an extend to the existing financial services of the company – “Create a mobile payment application”

Then, you proceed with the high-level requirements. They are called EPICs:

- EPIC example: “Creation of an electronic wallet”; “Deposit funds to e-wallet”; “Transfer funds to recipients”;

EPICS are still high-level so we move down to the details by defining “User Stories”. They are detailed descriptions of what the customer or user of the product wants to be able to do. Their main goal is to translate the user expectations, desires, wishes, into separate easy-to understand requirements or “stories”.

- User Story example: “As a user, I want to create wallet via registering through e-mail; “As a user, I want to be able to top-up my wallet by using Bank account”;

The User stories finally give the needed granularity. However, one more step is needed for the developers – the User Stories need to be translated into technical tasks. They are called “Features”:

Roles and responsibilities

Scrum Master responsibilities:

- Leads the Scrum team
- Guides the Product Owner in defining the backlog priorities
- Works to ensure the Sprint commitments are met
- Works to remove roadblocks for the development team
- Facilitates Daily Stand-Up meetings (DSUs)
- Facilitates Demo Sessions
- Facilitates Backlog grooming and Retrospective sessions



Scrum Master

Development (or delivery) team



Development team responsibilities:

- Perform work as per backlog priority
- Deliver ready to use product each Sprint
- Work with customer to translate user stories into features
- Develop software features according to priority
- Resolve technical issues and test solutions

Product Owner responsibilities:

- Is or represents the product's customer or user
- Defines the backlog
- Reviews and accepts the completed backlog
- Prioritizes the backlog for each Sprint

Product Owner

Everyone needs to Collaborate

Roles and responsibilities

Scrum Master responsibilities:

- Leads the Scrum team
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EVERYONE NEEDS TO COLLABORATE

Product owner responsibilities:

- Is or represents the product's customer or user
- Defines the backlog
- Reviews and accepts the completed backlog
- Prioritizes the backlog for each Sprint
- Approves/requests changes to the backlog priority

Development team responsibilities:

- Perform work as per backlog priority
- Deliver ready to use product each Sprint
- Work with customer to translate user stories into features
- Develop software features according to priority
- Resolve technical issues and test solutions

SCRUM: Project overview (e.g. Software development)

The standard phases of software development include Designing the solution, Coding and finally – Testing the software. The original way of structuring such similar work included performing these steps only once (designing the whole work, coding the whole program and only then – testing the entire software created).

This has been leading very often to heavy reworks and multiple changes in a single project.



The lifecycle of a Scrum project suggests, on the other hand, to work step by step – designing, coding and testing one part, before moving to the next one:

Overall Planning (Sprint 0)

Sprint 1:

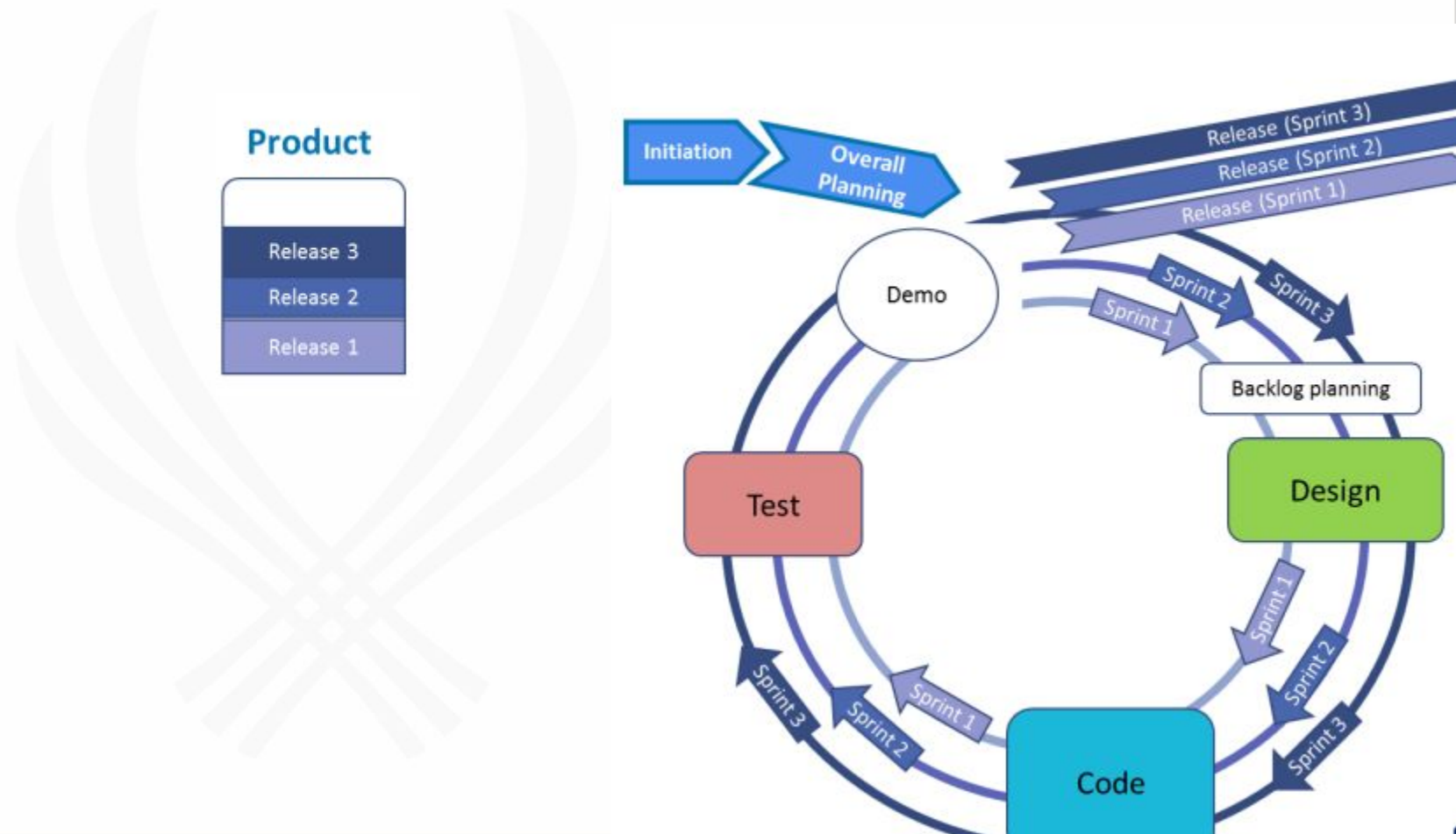
- Backlog Planning
- Execute work
- Demo
- Release of completed product

Sprint 2:

- Backlog Planning
- Execute
- Demo
- Release of completed product

Sprint 3: (...)

In order to ensure the work meets expectations and needs, the Product Owner has to play a very active role. The PO has to always attend the Planning and Demo sessions of each Sprint.



User story template

- <https://docs.google.com/spreadsheets/d/1Be7j8iUMkOoLqofL102YvYBFXpK7zNkh/edit?usp=sharing&ouid=107431048622011300376&rtpof=true&sd=true>

#story — unique story number.

story_title — short name of the user story.

story description — what the user wants to accomplish and why.

story_estimation / range out of __ — estimated effort, e.g., 5/10 or 8/13.

HC&HV — High Complexity & High Value.

LC&HV — Low Complexity & High Value.

HC&LV — High Complexity & Low Value.

LC&LV — Low Complexity & Low Value.

User_role — e.g., Student, Teacher, Administrator, Customer.

Pre-Requirements — conditions that must be completed before starting the story.

QA Test — how the story will be tested/verified.

Responsible person in a team — team member responsible for implementation.

Iteration — Sprint/Iteration number.

Deadline — planned completion date.

Materials

- Week 1 Materials:
- <https://drive.google.com/drive/folders/17isFKpsD6FCw18yh9Gy3O00PRB2-SfPu>
- Course Materials:
- https://drive.google.com/drive/folders/16K14MloBo7IErvVuj0ao7IM2xSVVcOf_

Questions

